Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

An atom of mass m is moving with a velocity v. It absorbs a photon of wavelength λ and comes to rest. The value of v is given by (here h is Planck's constant).

(A) $\frac{m\lambda}{h}$

(B) $\frac{mh}{\lambda}$

(C) $\frac{h}{m\lambda}$

(D) $\frac{\lambda h}{m}$

Q.2

$9\ \mathrm{kW}$ light source emits light of frequency $10^{15}\ \mathrm{Hz}$. A parallel beam of this light falls normally on a surface. If the surface completely absorbs all the incident photons, the force exerted by the light beam on the surface is

(A) $3 \times 10^{10}\ \mathrm{N}$

(B) $3 \times 10^{5}\ \mathrm{N}$

(C) $3 \times 10^{-5}\ \mathrm{N}$

(D) $3 \times 10^{-9}\ \mathrm{N}$

Q.3

An electromagnetic radiation given by
$E = E_0\left[\sin\left(2\pi v_1 t\right) + \sin\left(2\pi v_2 t\right)\right]$
where $v_1 = 1.3 \times 10^{15}\ \mathrm{Hz}$ and $v_2 = 1.1 \times 10^{15}\ \mathrm{Hz}$ falls on a photocell whose work function is $2.0\mathrm{eV}$. Then the photoelectric current through the cell

(A) is zero

(B) increases with time

(C) remains constant

(D) increases and decreases periodically with time

Q.4

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When a photon of light falls on a metal, it is absorbed, resulting in the emission of a photoelectron. Before the photon is completely absorbed by the metal, it penetrates a distance of about

(A) $1\ \text{mm}$

(B) $10^{-4}\ \text{m}$

(C) $10^{-8}\ \text{m}$

(D) $10^{-14}\ \text{m}$

Q.5

Choose the correct statement. The work function

(A) is the same for all metals

(B) is different for different metals

(C) is small for heavy elements

(D) depends upon the frequency of the incident radiation.

Q.6

When a point source of light is at a distance of $50\ \text{cm}$ from a photoelectric cell, the cut-off voltage is found to be V_0. If the same source is placed at a distance of $1\ \text{m}$ from the cell, the cut-off voltage will be

(A) $V_0/4$

(B) $V_0/2$

(C) V_0

(D) $2V_0$

Q.7

No photoelectrons are emitted from a metal if the wavelength of light exceeds $6000\tilde{A}$. The work function of the metal is approximately equal to

(A) $3 \times 10^{-16}\ \text{J}$

(B) $3 \times 10^{-19}\ \text{J}$

(C) $3 \times 10^{-20}\ \text{J}$

(D) $3 \times 10^{-22}\ \text{J}$

Q.8

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The threshold wavelength for photoelectric emission from a material is $5200\mathring{A}$ . . .. Photoelectrons will be emitted when this material is illuminated with monochromatic radiation from a

(A) 50 -watt infrared lamp

(B) 100 -watt red neon lamp

(C) 40-watt sodium lamp

(D) 5 -watt ultraviolet lamp

Q.9

The wavelength in air associated with a photon of energy E is (c is the speed of light in air and h is the Planck's constant)

(A) $\frac{hc}{E}$

(B) $\frac{h}{cE}$

(C) $\frac{c}{hE}$

(D) hcE

Q.10

A metal whose work function is $3.3\mathrm{eV}$ is illuminated by light of wavelength 3×10^{-7} m. What is the threshold frequency for photoelectric emission? Planck's constant $= 6.6 \times 10^{-34}\mathrm{Js}$

(A) 0.4×10^{15} Hz

(B) 0.8×10^{15} Hz

(C) 1.6×10^{15} Hz

(D) 3.2×10^{15} Hz

Q.11

X-rays of frequency v are used to irradiate sodium and copper surface in two separate experiments and the stopping potential determined. Then

(A) the stopping potential is more for copper than for sodium.

(B) the stopping potential is more for sodium than for copper.

(C) the stopping potential is the same for sodium and copper.

(D) the stopping potential for both will vary as $1/v$.

Q.12

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When monochromatic light falls on a photosensitive material, the number of photoelectrons emitted per second is n and their maximum kinetic energy is $K_{\max}$. If the intensity of the incident light is doubled, then

(A) both n and $K_{\max}$ are doubled

(B) both n and $K_{\max}$ are halved

(C) n is doubled but $K_{\max}$ remains the same

(D) $K_{\max}$ is doubled but n remains the same.

Q.13

The momentum of a photon of wavelength λ is

(A) $\frac{h}{c\lambda}$

(B) $\frac{hc}{\lambda}$

(C) $h\lambda$

(D) $\frac{h}{\lambda}$

Q.14

The energy of a photon corresponding to the visible light of maximum wavelength is approximately equal to

(A) 1.0eV

(B) 1.5eV

(C) 2.0eV

(D) 2.5eV

Q.15

When a certain photosensitive surface is illuminated with monochromatic light of frequency v, the stopping potential for photoelectric current is $V_0/2$. When the same surface is illuminated by monochromatic light of frequency $v/2$, the stopping potential is V_0. The threshold frequency for photoelectric emission is

(A) $\frac{2v}{3}$

(B) $\frac{3v}{2}$

(C) $\frac{3v}{5}$

(D) $\frac{5v}{3}$

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Q.16

When a certain photosensitive surface is illuminated with monochromatic light of wavelength λ, the stopping potential for photoelectric current is $2V_0$. When the same surface is illuminated with monochromatic light of wavelength 2λ, the stopping potential is $V_0/2$. The threshold wavelength for photoelectric emission is

(A) 1.5λ

(B) 2.0λ

(C) 2.5λ

(D) 3λ

Q.17

The threshold frequency for a certain photosensitive metal is v_0. When it is illuminated by light of frequency $v = 2v_0$, the maximum velocity of photoelectrons is v_0. What will be the maximum velocity of the photoelectrons when the same metal is illuminated by light of frequency $v = 5v_0$?

(A) $\sqrt{2}v_0$

(B) $2v_0$

(C) $2\sqrt{2}v_0$

(D) $4v_0$

Q.18

Lights of two different frequencies, whose photons have energies 2eV and 10eV respectively, successively illuminate a metal of work function 1eV. The ratio of the maximum speeds of the emitted electrons will be

(A) $1:5$

(B) $3:11$

(C) $1:9$

(D) $1:3$

Q.19

The kinetic energies of photoelectrons emitted from a metal are K_1 and K_2 when it is irradiated with lights of wavelength λ_1 and λ_2 respectively. The work function of the metal is

(A) $\frac{K_1\lambda_1 - K_2\lambda_2}{\lambda_2 - \lambda_1}$

(B) $\frac{K_1\lambda_1 + K_2\lambda_2}{\lambda_2 + \lambda_1}$

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(C) $\frac{K_1\lambda_2-K_2\lambda_1}{\lambda_2-\lambda_1}$

(D) $\frac{K_1\lambda_2+K_2\lambda_1}{\lambda_2+\lambda_1}$

Q.20

In an electron microscope, electrons accelerated by a voltage of 50kV are used. In an optical microscope of the same aperture, yellow light of wavelength $590\ \text{nm}$ is used. The resolving power of the electron microscope will be n times higher than that of the optical microscope, where n is of the order of

(A) 10^2

(B) 10^3

(C) 10^4

(D) 10^5

Q.21

The maximum wavelength of radiation that can cause photoelectric emission in a metal of work function 2.5eV is (in nm)

Q.22

Electrons accelerated by potential V are diffracted from a crystal. If $d = 1\tilde{A}\ldots$ and $i = 30^\circ, V$ should be about $\left(h = 6.6 \times 10^{-34}\text{Js}, \text{m} = 9.1 \times 10^{-31}\ \text{kg}\ \ e = 1.6 \times 10^{-19}\text{C}\right)$ (in Volts)

Q.23

An X-ray tube produces a continuous spectrum of radiation with its short-wavelength end at $0.33\tilde{A}\ldots$. What is the maximum energy of a photon in the radiation? Planck's constant $= 6.6 \times 10^{-34}\text{Js}$ (in eV)

Q.24

The work function for aluminium surface is $4.125\ \text{eV}$. The cut off wavelength for photoelectric effect for this surface is _____ (in Angstrom). Take $h = 6.6 \times 10^{-34}$

Q.25

10^{20} photons of wavelength $660\ \text{nm}$ are emitted per second from a lamp. What is the wattage of the lamp. Planck's constant $= 6.6 \times 10^{-34}\text{Js}$ (in Watt)

Q.26

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The work function of a substance is $4.0\mathrm{eV}$. The longest wavelength of light that can cause photoelectron emission from this substance is approximately (in nm)

Q.27

The maximum kinetic energy of photoelectrons emitted from a metal when photons of energy $10\mathrm{eV}$ fall on it is $3\mathrm{eV}$. The stopping potential is (in Volt)

Q.28

Davisson and Germer experiment is performed first with X-rays of wavelength $1\tilde{A}$ and then with electrons of wavelength $1\tilde{A}$ The ratio of energy of X-rays to that of the electrons is of the order of $(m =$ mass of electron)

Q.29

The work functions of silver and sodium are 4.6 and 2.3 eV, respectively. The ratio of the slope of the stopping potential versus frequency plot for silver to that of sodium is

Q.30

A photocell is illuminated by a small bright source placed $1\ \mathrm{m}$ away. When the same source is placed $\frac{1}{2}\ \mathrm{m}$ away, the number of photoelectrons emitted becomes _____ times.

Answers & Solutions

Q.1 Answer:

$\frac{h}{m\lambda}$

Solution:

The linear momentum of the photon is

$p = \frac{h}{\lambda}$

Since the atom comes to rest after absorbing the photon, the total final momentum is zero. From the principle of conservation of momentum, the initial momentum must be zero. Hence

$$mv = \frac{h}{\lambda}$$

$$\Rightarrow \quad v = \frac{h}{m\lambda}, \text{ which is choice (c).}$$

Q.2 Answer:

$3 \times 10^{-5}\ \mathrm{N}$

Solution:

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Total electromagnetic energy falling on the surface per second is

$E = 9 \times 10^3$ J per second

The total momentum of all the photons falling on the surface per second is

$p = \frac{E}{c} = \frac{9 \times 10^3}{3 \times 10^8} = 3 \times 10^{-5}$Ns

Since all the photons falling on the surface are completely absorbed, the total momentum imparted to the surface per second $= 3 \times 10^{-5}$Ns. According to Newton's second law, the change in momentum in 1 s is the force. Hence, the force exerted on the surface is

$F = \frac{3 \times 10^{-5}\text{Ns}}{1\text{ s}} = 3 \times 10^{-5}$ N

Q.3 Answer:

increases and decreases periodically with time

Solution:

Using trigonometric relation

$\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$

the above equation can be written as

$E = 2E_0 \cos\left\{2\pi\left(\frac{v_1 - v_2}{2}\right)t\right\} \sin\left\{2\pi\left(\frac{v_1+v_2}{2}\right)t\right\}$

$\Rightarrow \quad E = R \sin(2\pi v_a t)$ **(i)**

where $R = 2E_0 \cos\left\{2\pi\left(\frac{v_1 - v_2}{2}\right)t\right\}$ **(ii)**

and $v_a = \frac{1}{2}(v_1 + v_2)$

It follows from Eq. (ii) that the amplitude of the resultant wave varies periodically at beat frequency $v_\text{b} = v_1 - v_2$. Hence the resultant intensity rises and falls periodically at this frequency. Since photoelectric current is directly proportional to the intensity of incident radiation, the correct choice is (d).

Q.4 Answer:

10^{-8} m

Solution:

The correct choice is (c).

Q.5 Answer:

is different for different metals

Solution:

The photon penetrates a distance of about 10^{-8} m before it is absorbed. It imparts all its energy to a single electron. Part of this energy is used up in freeing the electron from the atom of the metal.

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This energy is called the work function which is a characteristic of the metal. Hence the correct choice is (b).

Q.6 Answer:

V_0

Solution:

When the distance of the source from the cell is increased, the intensity of light falling on the cell decreases; but the cut-off voltage of the cell is independent of the intensity of the incident radiation. Hence the correct choice is (c).

Q.7 Answer:

3×10^{-19} J

Solution:

Cut-off wavelength $\lambda_0 = 6000\tilde{A} \ldots = 6 \times 10^{-7}$ m **Therefore, threshold frequency is**
$v_0 = \frac{c}{\lambda_0} = \frac{3 \times 10^8}{6 \times 10^{-7}} = 0.5 \times 10^{15}$ Hz
Work function $W_0 = hv_0 = 6.6 \times 10^{-34} \times 0.5 \times 10^{15}$
$= 3.3 \times 10^{-19}$ J

Q.8 Answer:

5 -watt ultraviolet lamp

Solution:

Photoelectrons will be emitted if the wavelength of light is less than the threshold wavelength $5200\tilde{A} \ldots$ **. Wavelength of sodium light is about** $6000\tilde{A} \ldots$**, that of red light is about** $8000\tilde{A} \ldots$ **and that of infrared light is about** $10,000\tilde{A} \ldots$ **. But the wavelength of ultraviolet light is less than** $2000\tilde{A} \ldots$ **. Hence the correct choice is (d).**

Q.9 Answer:

$\frac{hc}{E}$

Solution:

Energy of a photon is $E = hv$**.**
Now $v = \frac{c}{\lambda}$**.**
Hence $E = \frac{hc}{\lambda}$
$\lambda = \frac{hc}{E}$
Hence the correct choice is (a).

Q.10 Answer:

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0.8×10^{15} Hz

Solution:

Work function $W_0 = hv_0 = 3.3\text{eV} = 3.3 \times 1.6 \times 10^{-19}$ J.

Therefore, the threshold frequency is

$$v_0 = \frac{W_0}{h} = \frac{3.3 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}}$$
$$= 0.8 \times 10^{15} \text{ Hz}$$

Q.11 Answer:

the stopping potential is more for sodium than for copper.

Solution:

Since the work function of sodium is lower than that of copper, it is easier to extract electrons from sodium than from copper. Therefore, the electrons ejected from sodium will have a greater kinetic energy and will hence need a greater stopping potential. Thus the correct choice is (b).

Q.12 Answer:

n is doubled but $K_{\max}$ remains the same

Solution:

The correct choice is (c).

Q.13 Answer:

$\frac{h}{\lambda}$

Solution:

Energy of photon $hv = mc^2$**. Therefore,**

$m = \frac{hv}{c^2} = \frac{h}{c\lambda} \quad (\because c = v\lambda)$

$\therefore$ **Momentum of photon** $= mc = \frac{h}{\lambda}$.

Hence the correct choice is (d)

Q.14 Answer:

1.5eV

Solution:

The maximum wavelength of visible light is about $8000\tilde{A}$ **. . . (the red end of the visible spectrum).**

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The energy of a photon of wavelength $\lambda = 8000\mathring{A} \ldots = 8 \times 10^{-7}$ m is

$$E = hv = \frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^{8}}{8 \times 10^{-7}}$$
$$\simeq 2.5 \times 10^{-19} \text{ J} \simeq 1.5\text{eV}$$

Q.15 Answer:

$\frac{3v}{2}$

Solution:

$hv = hv_0 + \text{e}V$, where V is the stopping potential. For frequency v, we have

$hv = hv_0 + \frac{eV_0}{2}$ (i)

and for frequency $\text{v}/2$, we have

$\frac{hv}{2} = hv_0 + eV_0$ (ii)

From (i) and (ii) on eliminating V_0, we get $v_0 = 3v/2$. Hence the correct choice is (b).

Q.16 Answer:

3λ

Solution:

$hv = hv_0 + eV$. Since $v = c/\lambda$, we have $\frac{hc}{\lambda} = \frac{hc}{\lambda_0} + eV$ where λ_0 is the threshold wavelength. For wavelength λ, we have

$\frac{hc}{\lambda} = \frac{hc}{\lambda_0} + 2eV_0$

and for wavelength 2λ, we have

$\frac{hc}{2\lambda} = \frac{hc}{\lambda_0} + \frac{eV_0}{2}$

Eliminating V_0 from (i) and (ii) we get $\lambda_0 = 3\lambda$ Hence the correct choice is (d).

Q.17 Answer:

$2v_0$

Solution:

$\frac{1}{2}mv_{\max}^2 = h\left(v - v_0\right)$. For light of frequency $v = 2v_0$ we have

$\frac{1}{2}mv_0^2 = h\left(2v_0 - v_0\right) = hv_0$ (i)

For light of frequency $v = 5v_0$, we have

$\frac{1}{2}mv^2 = h\left(5v_0 - v_0\right) = 4hv_0$ (ii)

Dividing (ii) by (i) we get $v^2 = 4v_0^2$ or $v = 2v_0$. Hence the correct choice is (b).

Q.18 Answer:

$1 : 3$

Solution:

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$E = \frac{1}{2}mv^2 = hv_0 - W_0$. **Now** $E_1 = 2 - 1 = 1\text{eV}$ **and** $E_2 = 10 - 1 = 9\text{eV}$.
Therefore $E_1/E_2 = 1/9$,
i.e. $\frac{\frac{1}{2}mv_1^2}{\frac{1}{2}mv_2^2} = \frac{1}{9}$
or $\frac{v_1}{v_2} = \frac{1}{3}$. **Hence the correct choice is (d).**

Q.19 Answer:

$\frac{K_1\lambda_1 - K_2\lambda_2}{\lambda_2 - \lambda_1}$

Solution:

$$\begin{aligned} &\text{Given} \quad \frac{hc}{\lambda_1} - W_0 = K_1 \quad \text{(i)} \\ &\text{and} \quad \frac{hc}{\lambda_2} - W_0 = K_2 \quad \text{(ii)} \end{aligned}$$

Eliminate hc **from (i) and (ii). The correct choice is (a).**

Q.20 Answer:

10^5

Solution:

$$\begin{aligned} \lambda_e &= \frac{h}{\sqrt{2meV}} \\ &= \frac{6.6 \times 10^{-34}}{\sqrt{2 \times (9 \times 10^{-31}) \times (1.6 \times 10^{-19}) \times (50 \times 10^3)}} \\ &\simeq 5.5 \times 10^{-12}\ \text{m} \\ \lambda_y &= 590\ \text{nm} = 5.9 \times 10^{-7}\ \text{m} \end{aligned}$$

Now, resolving power $\propto \frac{1}{\lambda}$. **Therefore**

$$\begin{aligned} \frac{\text{R.P.of electron microscope}}{\text{R.P.of optical microscope}} &= \frac{\lambda_y}{\lambda_e} \\ &= \frac{5.9 \times 10^{-7}}{5.5 \times 10^{-12}} \\ &\simeq 10^5 \end{aligned}$$

So the correct choice is (d).

Q.21 Answer:

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Solution:

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$$\lambda_{\max} = \frac{hc}{W_0} = \frac{\left(6.63 \times 10^{-34}\right) \times \left(3 \times 10^{8}\right)}{\left(2.5 \times 1.6 \times 10^{-19}\right)}$$
$$= 4.97 \times 10^{-7}\ \text{m}$$
$$= 497\ \text{nm}$$

Q.22 Answer:

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Solution:

For constructive interference, path difference $\Delta = n\lambda$.

Path difference $\Delta = QP + PR$
$$= OP\cos i + OP\cos i$$
$$= 2d\cos i$$
$$\therefore \quad 2d\cos i = n\lambda \Rightarrow \lambda = \frac{2d\cos i}{n}$$
$$= \frac{2 \times 1\tilde{A}\ldots \times \cos 30^\circ}{n} = \frac{\sqrt{3}}{n}\tilde{A}\ldots$$

For $n = 1, \lambda = \sqrt{3}\tilde{A}\ldots = \sqrt{3} \times 10^{-10}$ m

Kinetic energy of an electron acceleration through a potential difference V is $K = eV$. de Broglie wavelength of electron is

$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2meV}}$

Putting $h = 6.6 \times 10^{-34}$Js, $m = 9.1 \times 10^{-31}$ kg **and** $e = 1.6 \times 10^{-19}$C, **we get**

$$\lambda = \frac{12.26 \times 10^{-10}}{\sqrt{V}}$$
$$\Rightarrow \quad \sqrt{3} \times 10^{-10} = \frac{12.26 \times 10^{-10}}{\sqrt{V}}$$

which gives $V = 50$ V

Q.23 Answer:

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Solution:

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$$E_{\max} = h\nu_{\max} = \frac{hc}{\lambda_{\min}} = \frac{6.6 \times 10^{-34} \times 3 \times 10^{8}}{0.33 \times 10^{-10}}$$
$$= 6 \times 10^{-15}\ \text{J} = 37.5\text{keV}$$

Q.24 Answer:

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Solution:

The cut-off wavelength λ_0 is given by

$$\lambda_0 = \frac{hc}{W_0} = \frac{6.6 \times 10^{-34} \times 3 \times 10^{8}}{4.125 \times 1.6 \times 10^{-19}}$$
$$= 3000 \times 10^{-10}\ \text{m} = 3000\tilde{A} \ldots$$

Q.25 Answer:

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Solution:

Energy of photons of frequency $(v) = hv$. If the lamp emits v photons per second, then the power of the lamp is $P = nhv$. Now, $v = c/\lambda$. Therefore

$$P = \frac{nhc}{\lambda}$$
$$= \frac{10^{20} \times 6.6 \times 10^{-34} \times 3 \times 10^{8}}{660 \times 10^{-9}} = 30\ \text{W}$$

Q.26 Answer:

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Solution:

$\lambda_{\max} = \frac{hc}{W_0} = \frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{4.0 \times 1.6 \times 10^{-19}} = 3.10 \times 10^{-7}\ \text{m}$
$= 310\ \text{nm}$

Q.27 Answer:

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Solution:

Stopping potential in volts = kinetic energy of the emitted photoelectrons in eV.

Q.28 Answer:

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Solution:

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$$E_X = h\nu = \frac{hc}{\lambda}$$

$$E_e = \frac{1}{2}mv^2 = \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2} \quad \left(\because \lambda = \frac{h}{p}\right)$$

$$\therefore \quad \frac{E_X}{E_e} = \frac{hc}{\lambda} \times \frac{2m\lambda^2}{h^2} = \frac{2mc\lambda}{h}$$

Q.29 Answer:

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Solution:

$ev_s = h\nu$ - hv_0

$v_s = \frac{h}{e}\nu$ - $\frac{h}{e}v_0$

⇒ **Slope is independent of stopping potential then ratio is 1 : 1.**

Q.30 Answer:

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Solution:

Intensity obeys inverse square law of distance, i.e. $I \propto \frac{1}{r^2}$**. Hence if** r **is halved, the intensity becomes 4 times. Also the number of photoelectrons emitted is proportional to the intensity of incident radius.**

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